Air University Islamabad Department of Computer Science

LAB-5

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Web Technologies Lab - Sir. Mohabbat Ali

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Lab Title: JavaScript Basics – Conditions, Loops, Functions, Object and Arrays
Objective: To understand JavaScript syntax, conditional statements, loops, functions, and arrays using HTML and `<script>`.

LAB ACTIVITY

Activity 1:

Add Two Numbers
To perform arithmetic operations using JavaScript and display the result on a webpage.

Concept Used

- Variables (let)
- Type conversion using Number()
- DOM manipulation
- Arithmetic operators

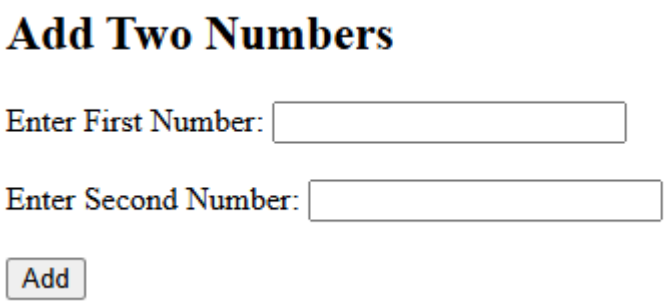
Explanation
This program takes two numbers from the user and adds them when the button is clicked. The result is displayed dynamically on the webpage.

CODE:

```
<!DOCTYPE html>
<html>
<head>
    <title>Add Two Numbers</title>
</head>
<body>

<h2>Add Two Numbers</h2>

<label>Enter First Number:</label>
<input type="number" id="num1"><br><br>

<label>Enter Second Number:</label>
<input type="number" id="num2"><br><br>

<button onclick="addNumbers()">Add</button>

<h3 id="result"></h3>

<script>
function addNumbers() {
    let n1 = Number(document.getElementById("num1").value);
    let n2 = Number(document.getElementById("num2").value);

    let sum = n1 + n2;
```

```
    document.getElementById("result").innerHTML = "Sum = " + sum;
}
</script>

</body>
</html>
```

OUTPUT



Activity 2:

Login System
To simulate a simple login validation.

Concept Used

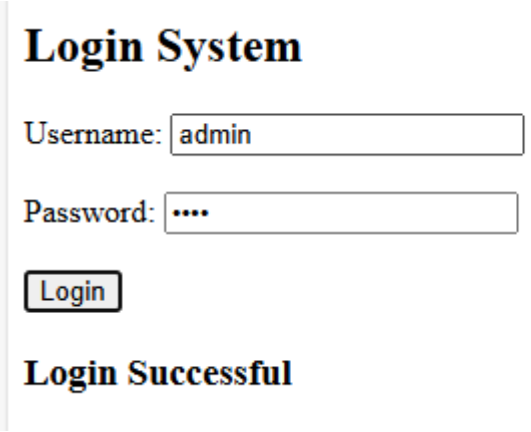
- Logical AND (&&)
- String comparison
- Conditional statements

CODE:

```
<!DOCTYPE html>
<html>
<head>
    <title>Login System</title>
</head>
<body>

<h2>Login System</h2>

<label>Username:</label>
<input type="text" id="username"><br><br>

<label>Password:</label>
<input type="password" id="password"><br><br>

<button onclick="login()">Login</button>
```

```
<h3 id="message"></h3>

<script>
function login() {
    let user = document.getElementById("username").value;
    let pass = document.getElementById("password").value;

    if (user === "admin" && pass === "1234") {
        document.getElementById("message").innerHTML = "Login Successful";
    } else {
        document.getElementById("message").innerHTML = "Invalid Credentials";
    }
}
</script>

</body>
</html>
```

OUTPUT



Explanation
The system validates predefined username and password and displays success or failure message.

Activity 3:

Array
Learn how to create and manipulate arrays in JavaScript.

Problem
Create an array of 5 student names. Perform the following:

- Display all names
- Add one new student

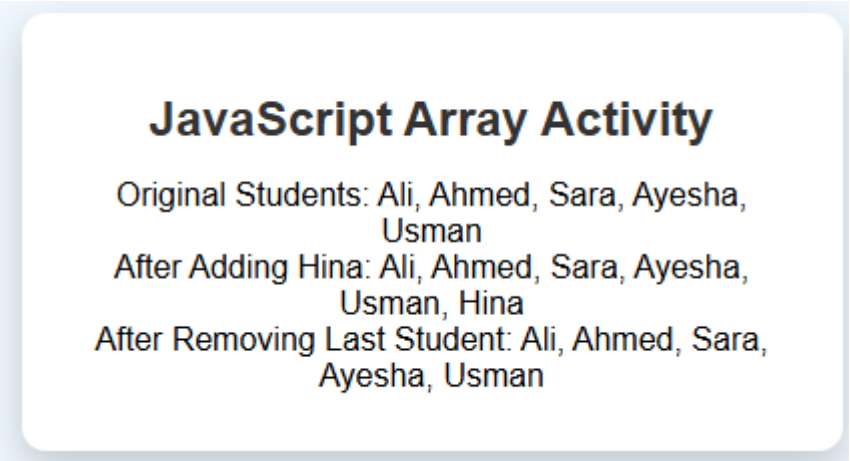
- Remove the last student
- Display the updated array

CODE:

```
<!DOCTYPE html>
<html>
<head>
    <title>JavaScript Array Activity</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f0f8ff;
            display: flex;
            justify-content: center;
            align-items: center;
            height: 100vh;
        }
        .container {
            background-color: #ffffff;
            padding: 20px 30px;
            border-radius: 12px;
            box-shadow: 0 8px 20px rgba(0,0,0,0.2);
            width: 350px;
            text-align: center;
        }
        h2 {
            color: #333;
            margin-bottom: 15px;
        }
        p {
            font-size: 16px;
            margin: 8px 0;
        }
    </style>
</head>
<body>

<div class="container">
    <h2>JavaScript Array Activity</h2>
    <p id="output"></p>
</div>

<script>
    // Step 1: Create array
    let students = ["Ali", "Ahmed", "Sara", "Ayesha", "Usman"];

    // Display original array
```

```
    let outputText = "Original Students: " + students.join(", ") + "<br>";

    // Step 2: Add new student
    students.push("Hina");
    outputText += "After Adding Hina: " + students.join(", ") + "<br>";

    // Step 3: Remove last student
    students.pop();
    outputText += "After Removing Last Student: " + students.join(", ");

    // Step 4: Display array
    document.getElementById("output").innerHTML = outputText;
</script>

</body>
</html>
```

OUTPUT



Activity 4:

Object
Understand how to create and access objects.

Problem
Create a student object with:

- name
- age
- department
- Display the student information.

CODE:

```html
<!DOCTYPE html>
<html>
<head>
    <title>Object Activity</title>
</head>
<body>

<h2>JavaScript Object Activity</h2>

<p id="info"></p>

<script>
// Step 1: Create object
let student = {
    name: "Ali",
    age: 21,
    department: "Computer Science"
};

// Step 2: Display object data
document.getElementById("info").innerHTML =
    "Name: " + student.name + "<br>" +
    "Age: " + student.age + "<br>" +
    "Department: " + student.department;
</script>

</body>
</html>
```

OUTPUT

JavaScript Object Activity

Name: Ali
Age: 21
Department: Computer Science

LAB TASK

TASK 1

Basic Calculator Enhancement
Modify Activity 1 (Add Two Numbers):

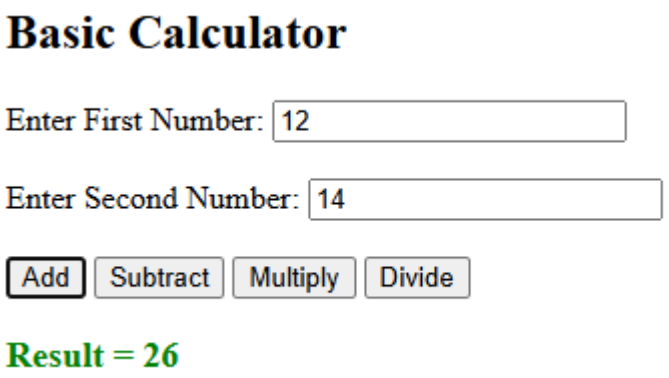
1. Add buttons for:
    - Subtraction
    - Multiplication
    - Division
2. Display an error message if user leaves input empty.
3. Change result color dynamically (green for valid result, red for error).

CODE:

```
<!DOCTYPE html>
<html>
<head>
    <title>Basic Calculator Enhancement</title>
    <style>
        #result {
            font-weight: bold;
            font-size: 18px;
        }
    </style>
</head>
<body>

<h2>Basic Calculator</h2>

<label>Enter First Number:</label>
<input type="number" id="num1"><br><br>

<label>Enter Second Number:</label>
<input type="number" id="num2"><br><br>

<button onclick="calculate('+')">Add</button>
<button onclick="calculate('-')">Subtract</button>
<button onclick="calculate('*')">Multiply</button>
<button onclick="calculate('/')">Divide</button>

<h3 id="result"></h3>

<script>
function calculate(operator) {
    let n1 = document.getElementById("num1").value;
    let n2 = document.getElementById("num2").value;
```

```
    let result = document.getElementById("result");

    if (n1 === "" || n2 === "") {
        result.innerHTML = "Error: Please enter both numbers.";
        result.style.color = "red";
        return;
    }

    n1 = Number(n1);
    n2 = Number(n2);
    let output;

    if (operator === "+") {
        output = n1 + n2;
    }
    else if (operator === "-") {
        output = n1 - n2;
    }
    else if (operator === "*") {
        output = n1 * n2;
    }
    else if (operator === "/") {
        if (n2 === 0) {
            result.innerHTML = "Error: Cannot divide by zero.";
            result.style.color = "red";
            return;
        }
        output = n1 / n2;
    }

    result.innerHTML = "Result = " + output;
    result.style.color = "green";
}
</script>

</body>
</html>
```

OUTPUT



TASK 2

Number Checker

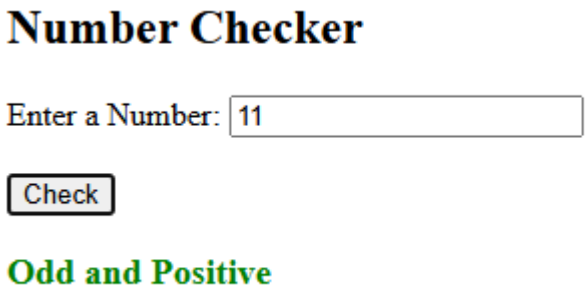
1. Check whether number is:
    - Even or Odd
2. Display message:
    - Green for positive
    - Red for negative
3. Show alert if input is empty.

CODE:

```
<!DOCTYPE html>
<html>
<head>
    <title>Number Checker</title>
    <style>
        #result {
            font-weight: bold;
            font-size: 18px;
        }
    </style>
</head>
<body>

<h2>Number Checker</h2>

<label>Enter a Number:</label>
<input type="number" id="number"><br><br>

<button onclick="checkNumber()">Check</button>
```

```html
<h3 id="result"></h3>

<script>
function checkNumber() {
    let num = document.getElementById("number").value;
    let result = document.getElementById("result");

    if (num === "") {
        alert("Please enter a number.");
        return;
    }

    num = Number(num);

    let type = (num % 2 === 0) ? "Even" : "Odd";

    if (num > 0) {
        result.style.color = "green";
        result.innerHTML = type + " and Positive";
    }
    else if (num < 0) {
        result.style.color = "red";
        result.innerHTML = type + " and Negative";
    }
    else {
        result.style.color = "black";
        result.innerHTML = "Zero (Even number)";
    }
}
</script>

</body>
</html>
```

OUTPUT



TASK 3

Array based web page
Create a web page using HTML, CSS, and JavaScript that performs the following:

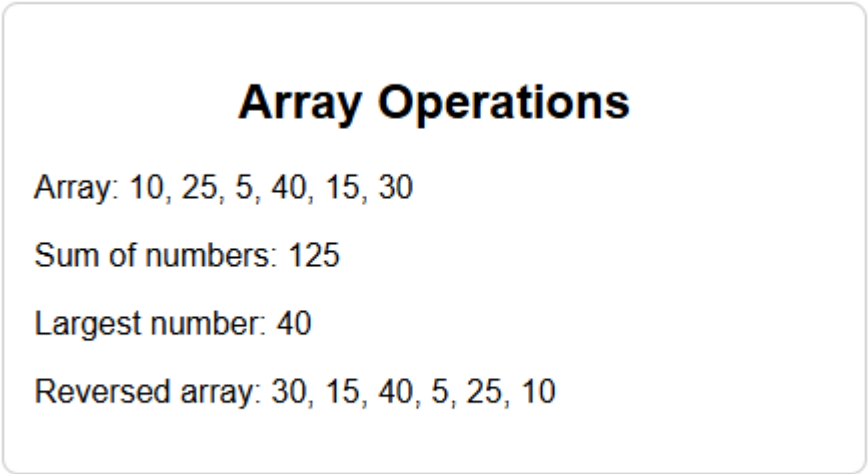
1. Create an array of 6 numbers.
2. Calculate and display the sum of all numbers.
3. Find and display the largest number in the array.
4. Display the array in reverse order.
5. Show all results on the web page (not in console).

CODE:

```html
<!DOCTYPE html>
<html>
<head>
    <title>Array Based Web Page</title>
    <style>
        body {
            font-family: Arial, sans-serif;
        }
        .container {
            width: 400px;
            margin: 20px auto;
            padding: 15px;
            border: 1px solid #ccc;
            border-radius: 8px;
        }
        h2 {
            text-align: center;
        }
        p {
            font-size: 16px;
        }
    </style>
</head>
<body>

<div class="container">
    <h2>Array Operations</h2>
    <p id="array"></p>
    <p id="sum"></p>
    <p id="largest"></p>
    <p id="reverse"></p>
</div>

<script>
    // 1. Create an array of 6 numbers
```

```
    let numbers = [10, 25, 5, 40, 15, 30];

    // Display original array
    document.getElementById("array").innerHTML =
        "Array: " + numbers.join(", ");

    // 2. Calculate sum
    let sum = 0;
    for (let i = 0; i < numbers.length; i++) {
        sum += numbers[i];
    }
    document.getElementById("sum").innerHTML =
        "Sum of numbers: " + sum;

    // 3. Find largest number
    let largest = numbers[0];
    for (let i = 1; i < numbers.length; i++) {
        if (numbers[i] > largest) {
            largest = numbers[i];
        }
    }
    document.getElementById("largest").innerHTML =
        "Largest number: " + largest;

    // 4. Display array in reverse order
    let reversed = [...numbers].reverse();
    document.getElementById("reverse").innerHTML =
        "Reversed array: " + reversed.join(", ");
</script>

</body>
</html>
```

OUTPUT



TASK 4

Array of Objects
Create a web page that:

1. Creates an array of 3 student objects.
   Each student must have:
   - name
   - marks
2. Display all students on the page.
3. Calculate and display the average marks.
4. Find and display the topper student.

Requirements

- Use loop over array of objects
- Display results in proper HTML format
- Apply CSS styling

CODE:

```
<!DOCTYPE html>
<html>
<head>
    <title>Array of Objects</title>
    <style>
        body {
            font-family: Arial, sans-serif;
            background-color: #f4f4f4;
```

```
        }
        .container {
            width: 500px;
            margin: 30px auto;
            padding: 20px;
            background-color: #ffffff;
            border-radius: 8px;
            box-shadow: 0 0 10px rgba(0,0,0,0.1);
        }
        h2 {
            text-align: center;
        }
        table {
            width: 100%;
            border-collapse: collapse;
            margin-bottom: 15px;
        }
        table, th, td {
            border: 1px solid #ccc;
        }
        th, td {
            padding: 8px;
            text-align: center;
        }
        .result {
            font-weight: bold;
            margin-top: 10px;
        }
        .topper {
            color: green;
        }
    </style>
</head>
<body>

<div class="container">
    <h2>Array of Student Objects</h2>

    <div id="studentsList"></div>
    <div id="average" class="result"></div>
    <div id="topper" class="result topper"></div>
</div>

<script>
    // 1. Create array of 3 student objects
    let students = [
        { name: "Ali", marks: 85 },
        { name: "Sara", marks: 92 },
```

```
        { name: "Ahmed", marks: 78 }
    ];

    let total = 0;
    let highest = students[0];

    // Create table
    let table = "<table><tr><th>Name</th><th>Marks</th></tr>";

    // 2. Display all students using loop
    for (let i = 0; i < students.length; i++) {
        table += "<tr><td>" + students[i].name + "</td><td>" +
students[i].marks + "</td></tr>";

        total += students[i].marks;

        if (students[i].marks > highest.marks) {
            highest = students[i];
        }
    }

    table += "</table>";
    document.getElementById("studentsList").innerHTML = table;

    // 3. Calculate and display average marks
    let average = total / students.length;
    document.getElementById("average").innerHTML =
        "Average Marks: " + average.toFixed(2);

    // 4. Find and display topper student
    document.getElementById("topper").innerHTML =
        "Topper: " + highest.name + " (" + highest.marks + " marks)";
</script>

</body>
</html>
```

OUTPUT

Array of Student Objects

Name	Marks
Ali	85
Sara	92
Ahmed	78

Average Marks: 85.00

Topper: Sara (92 marks)

---END---